Introduction to Mathematics and Optimisation

Exercises Week 10

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1 Exercise 7.39

Computing the distance from the line $y = x + 1$ to the point $(1, 1)$ gives rise to the minimization problem

$$\begin{array}{ll} \text{Minimize} & (x-1)^2 + (y-1)^2 \\ \text{with constraint} & y = x + 1. \end{array}$$

Solve this minimization problem using Theorem 7.37.

1.1 Solution

Using the method of Lagrange multipliers we get the following equations:

$$\begin{aligned} x - y + 1 &= 0 \\ 2x - 2 + \lambda &= 0 \\ 2y - 2 - \lambda &= 0 \end{aligned}$$

Then we solve this set of linear equations and we get $(x, y) = (0.5, 1.5)$ and $\lambda = 1$. Now we can compute the hessian matrix which yields $\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$. Since the hessian matrix is positive definite for all (x, y) the function is strictly convex by Theorem 8.23 and by theorem 6.13 the point we found has to be global minimum. And by theorem 8.12 since the hessian is positive definite at the point $(0.5, 1.5)$ the point is a minimum. Now we evaluate the function at point $(0.5, 1.5)$ which is equal to 0.5 and since this is distance squared the distance between the line $y = x + 1$ and point $(1, 1)$ is $\sqrt{0.5}$

2 Exercise 7.46

Solve the two optimization problems

$$\begin{array}{ll} \text{Maximize/Minimize} & x^2 - 2xy + 2y \\ \text{with constraint} & (x, y) \in C, \end{array}$$

where $C = \{(x, y) \in \mathbb{R}^2 \mid 0 \leq x \leq 3, 0 \leq y \leq 2\}$. But first give a reason as to why they both are solvable.

2.1 Solution

According to the Theorem 5.66 the problem has a solution if C is compact and the function we're trying to minimize is continuous. The subset C is a rectangle including it's own boundary therefore it is compact and the function is polynomial which is continuous by Remark 5.59.

2.1.1 Interior

According to proposition 7.44 if the solution is on the interior it's the critical point. To find the critical point we need to find where the gradient of the function is 0. The gradient of the function is $(2x - 2y, -2x + 2)$. Setting these to 0 we get $x = 1$ and $x = y = 1$. And at this point the function evaluates to 1. So the first candidate is $(1, 1)$ where $f(1, 1) = 1$

2.1.2 Boundary

The boundary consists of 4 lines segments. We optimize f on each of the segments.

Bottom Edge:

The bottom edge is $\partial C_b = \{(x, y) \in \mathbb{R} | y = 0, 0 \leq x \leq 3\}$. We can substitute $y = 0$ to the function receiving $f(x, 0) = x^2 - 0 + 0 = x^2$. And since x^2 is strictly increasing on the interval $[0, 3]$ we only need to check the boundary points.

- Min at $x = 0$: $f(0, 0) = 0$
- Max at $x = 3$: $f(3, 0) = 9$

Top Edge

The top edge is $\partial C_t = \{(x, y) \in \mathbb{R} | y = 2, 0 \leq x \leq 3\}$. We can substitute $y = 2$ to the function receiving $f(x, 0) = (x - 2)^2$. We check the critical point and boundaries:

- Critical point: $f(2, 2) = 0$
- $x = 0$: $f(0, 2) = 4$
- $x = 3$: $f(3, 2) = 1$

Left Edge

The left edge is $\partial C_l = \{(x,y) \in \mathbb{R} | 0 \leq y \leq 2, x = 0\}$. We can substitute $x = 0$ to the function receiving $f(x,0) = 2y$. Since this function is strictly increasing on the interval $[0,2]$ we just need to check the boundaries:

- Min at $y = 0$: $f(0,0) = 0$
- Max at $y = 2$: $f(0,2) = 4$

Right Edge

The right edge is $\partial C_r = \{(x,y) \in \mathbb{R} | 0 \leq y \leq 2, x = 3\}$. We can substitute $x = 0$ to the function receiving $f(x,0) = 9 - 4y$. This function is strictly increasing on the interval $[0,2]$ we just need to check the boundaries:

- Max at $y = 0$: $f(3,0) = 9$
- Min at $y = 2$: $f(3,2) = 1$

Now between all of these candidates we can select the minimum and maximum

- Maximum is at point $(3,0)$ with the value of 9
- Minimum is at points $(0,0)$ and $(2,2)$ with the value of 0